

# Duane S. Boning

professor

## 基本信息

姓名: Duane S. Boning

所属院校: 麻省理工学院

地区: United States

年龄范围: 50岁以上

企业经历: 有

## 荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可, 展现了卓越的研究能力和影响力。

## 学科领域

微电子科学与工程

光电信息科学与工程

集成电路设计与集成系统

## 研究领域

微电子器件与制造

EDA工具与电路验证

光电子器件与微纳光学

低功耗电路与能量管理

## 个人简介

Dr. Duane S. Boning is the Clarence J. LeBel Professor in Electrical Engineering and Professor of Electrical Engineering and Computer Science at MIT's EECS Department. He is affiliated with the MIT Microsystems Technology Laboratories and serves as MTL Associate Director for Computation and CAD. Dr. Boning is also the Engineering Faculty Co-Director of the MIT Leaders for Global Operations (LGO) program. His research focuses on the modeling and control of variation in IC, photonics, and MEMS processes, devices, and circuits,

with particular emphasis on statistical characterization and design for manufacturing. He has extensive teaching experience in programming, machine learning, and manufacturing process control.

## 工作经历

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**Massachusetts Institute of Technology - Clarence J. LeBel Professor in Electrical Engineering**

**Massachusetts Institute of Technology - Professor of Electrical Engineering and Computer Science**

**MIT Microsystems Technology Laboratories - MTL Associate Director for Computation and CAD**

**Massachusetts Institute of Technology - Engineering Faculty Co - Director of the MIT Leaders for Global Operations (LGO) program** 2016-09 - 至今

**Massachusetts Institute of Technology - Associate Head of the EECS Department** 2004 - 2011

**Massachusetts Institute of Technology - Director/Faculty Lead of the MIT Skoltech Initiative** 2011 - 2013

**Massachusetts Institute of Technology - faculty Director of the MIT/Masdar Institute Cooperative Program** 2011 - 2018-07

**Texas Instruments Semiconductor Process and Design Center - Member Technical Staff** 1991 - 1993

**IEEE Transactions on Semiconductor Manufacturing - Editor in Chief** 2001 - 2011

**Labnetwork mailing list - Moderator** -

## 教育经历

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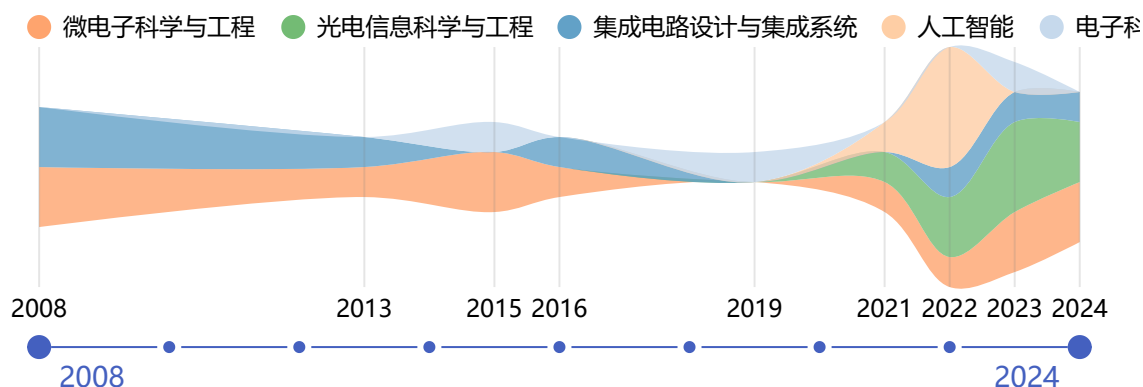
**Massachusetts Institute of Technology - S.B.** - 1984

**Massachusetts Institute of Technology - S.M.** - 1986

**Massachusetts Institute of Technology - Ph.D.** - 1991

## 近年发表的论文

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## Selecting robust silicon photonic designs after Bayesian optimization without extra simulations

OPTICS EXPRESS

Gao, Zhengqi; Zhang, Zhengxing; He, Zichang; Gu, Jiaqi; Pan, David Z.; Boning, Duane S.

被引用次数: 0

2024

## Bound-Constrained Expectation Maximization for Weibull Competing-Risks Device Reliability

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Chakraborty, Uttara; Boning, Duane S.; Thompson, Carl V.

被引用次数: 0

2024

## Modern Trends in Microelectronics Packaging Reliability Testing

MICROMACHINES

Bender, Emmanuel; Bernstein, Joseph B.; Boning, Duane S.

被引用次数: 1

2024

## Provable Routing Analysis of Programmable Photonic Circuits

JOURNAL OF LIGHTWAVE TECHNOLOGY

Gao, Zhengqi; Chen, Xiangfeng; Zhang, Zhengxing; Lai, Chih-Yu; Chakraborty, Uttara; Bogaerts, Wim; Boning, Duane S.

被引用次数: 0

2024

## Gradient-Based Power Efficient Functional Synthesis for Programmable Photonic Circuits

JOURNAL OF LIGHTWAVE TECHNOLOGY

Gao, Zhengqi; Chen, Xiangfeng; Zhang, Zhengxing; Chakraborty, Uttara; Bogaerts, Wim; Boning, Duane S.

被引用次数: 0

2024

## Identification of Multiple Failure Mechanisms for Device Reliability Using Differential Evolution

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Chakraborty, Uttara; Bender, Emmanuel; Boning, Duane S.; Thompson, Carl V.

被引用次数: 1

2023

## Automatic synthesis of light-processing functions for programmable photonics: theory and realization

PHOTONICS RESEARCH

Gao, Zhengqi; Chen, Xiangfeng; Zhang, Zhengxing; Chakraborty, Uttara; Bogaerts, Wim; Boning, Duane S.

被引用次数: 4

2023

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**Few-Shot Bayesian Performance Modeling for Silicon Photonic Devices Under Process Variation**

JOURNAL OF LIGHTWAVE TECHNOLOGY

Gao, Zhengqi; Zhang, Zhengxing; Boning, Duane S.

被引用次数: 1

2023

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**Inference of process variations in silicon photonics from characterization measurements**

OPTICS EXPRESS

Zhang, Zhengxing; El-Henawy, Sally I.; Ocampo, Carlos A. Rios; Boning, Duane S.

被引用次数: 1

2023

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**Impact of process variations on splitter-tree-based integrated optical phased arrays**

OPTICS EXPRESS

Zhang, Zhengxing; Notaros, Milica; Gao, Zhengqi; Chakraborty, Uttara; Notaros, Jelena; Boning, Duane S.

被引用次数: 4

2023

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**Intelligent Optimization of Dosing Uniformity in Ion Implantation Systems**

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Lang, Christopher, I; Sprenkle, Rick; Wilson, Eric; Samolov, Ana; Boning, Duane S.

被引用次数: 0

2022

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**Variational Quantum Pulse Learning**

2022 IEEE INTERNATIONAL CONFERENCE ON QUANTUM COMPUTING AND ENGINEERING (QCE 2022)

Liang, Zhiding; Wang, Hanrui; Cheng, Jinglei; Ding, Yongshan; Ren, Hang; Gao, Zhengqi; Hu, Zhirui; Boning, Duane S.; Qian, Xuehai; Han, Song; Jiang, Weiwen; Shi, Yiyu

被引用次数: 14

2022

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**Learning from Multiple Annotator Noisy Labels via Sample-Wise Label Fusion**

COMPUTER VISION, ECCV 2022, PT XXIV

Gao, Zhengqi; Sun, Fan-Keng; Yang, Mingran; Ren, Sucheng; Xiong, Zikai; Engeler, Marc; Burazer, Antonio; Wildling, Linda; Daniel, Luca; Boning, Duane S.

被引用次数: 9

2022

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**NeurOLight: A Physics-Agnostic Neural Operator Enabling Parametric Photonic Device Simulation**

ADVANCES IN NEURAL INFORMATION PROCESSING SYSTEMS 35 (NEURIPS 2022)

Gu, Jiaqi; Gao, Zhengqi; Feng, Chenghao; Zhu, Hanqing; Chen, Ray T.; Boning, Duane S.; Pan, David Z.

被引用次数: 0

2022

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**Mitigation of Thermal Stability Concerns in FinFET Devices**

ELECTRONICS

Bender, Emmanuel; Bernstein, Joseph B.; Boning, Duane S.

被引用次数: 0

2022

## Modeling and Optimizing the Impact of Process and Equipment Parameters in Sputtering Deposition Systems Using a Gaussian Process Machine Learning Framework

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Lang, Christopher I.; Jansen, Alexander; Didari, Sima; Kothnur, Prashanth; Boning, Duane S.

被引用次数: 5

2022

## Understanding and Improving Virtual Metrology Systems Using Bayesian Methods

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Lang, Christopher, I; Sun, Fan-Keng; Veerasingam, Ramana; Yamartino, John; Boning, Duane S.

被引用次数: 3

2022

## One Class Process Anomaly Detection Using Kernel Density Estimation Methods

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Lang, Christopher, I; Sun, Fan-Keng; Lawler, Bruce; Dillon, Jack; Al Dujaili, Ash; Ruth, John; Cardillo, Peter; Alfred, Perry; Bowers, Alan; Mckiernan, Adrian; Boning, Duane S.

被引用次数: 12

2022

## Variation-Aware Methods and Models for Silicon Photonic Design-for-Manufacturability

JOURNAL OF LIGHTWAVE TECHNOLOGY

Boning, Duane S.; El-Henawy, Sally, I; Zhang, Zhengxing

被引用次数: 4

2022

## Automatic Synthesis of Broadband Silicon Photonic Devices via Bayesian Optimization

JOURNAL OF LIGHTWAVE TECHNOLOGY

Gao, Zhengqi; Zhang, Zhengxing; Boning, Duane S.

被引用次数: 6

2022

## Enabling Wavelength-Dependent Adjoint-Based Methods for Process Variation Sensitivity Analysis in Silicon Photonics

JOURNAL OF LIGHTWAVE TECHNOLOGY

Zhang, Zhengxing; El-Henawy, Sally, I; Sadun, Allan; Miller, Ryan; Daniel, Luca; White, Jacob K.; Boning, Duane S.

被引用次数: 4

2021

## VARIATIONAL INFERENCE FORMULATION FOR A MODEL-FREE SIMULATION OF A DYNAMICAL SYSTEM WITH UNKNOWN PARAMETERS BY A RECURRENT NEURAL NETWORK

SIAM JOURNAL ON SCIENTIFIC COMPUTING

Yeo, Kyongmin; Grullon, Dylan E. C.; Sun, Fan-Keng; Boning, Duane S.; Kalagnanam, Jayant R.

被引用次数: 3

2021

## Modeling and Controlling Layout Dependent Variations in Semi-Additive Copper Electrochemical Plating

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Lang, Christopher, I; Boning, Duane S.

被引用次数: 2

2019

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**Modeling Spin Coating Over Topography and Uniformity Improvements Through Fill Patterns for Advanced Packaging Technologies**

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Lang, Christopher I.; Boning, Duane S.

被引用次数: 5

2019

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**Compact Model Parameter Extraction Using Bayesian Inference, Incomplete New Measurements, and Optimal Bias Selection**

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Yu, Li; Saxena, Sharad; Hess, Christopher; Elfadel, Ibrahim (Abe) M.; Antoniadis, Dimitri A.; Boning, Duane S.

被引用次数: 3

2016

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**Multistrata Subsurface Laser-Modified Microstructure With Background-Assisted Controlled Fracture for Defect-Free Ultrathin Die Fabrication**

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Teh, Weng Hong; Boning, Duane S.; Welsch, Roy E.

被引用次数: 5

2015

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**Multi-Strata Stealth Dicing Before Grinding for Singulation-Defects Elimination and Die Strength Enhancement: Experiment and Simulation**

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Teh, Weng Hong; Boning, Duane S.; Welsch, Roy E.

被引用次数: 21

2015

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**A 12b 250 MS/s Pipelined ADC With Virtual Ground Reference Buffers**

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Boo, Hyun H.; Boning, Duane S.; Lee, Hae-Seung

被引用次数: 33

2015

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**Efficient Spatial Pattern Analysis for Variation Decomposition via Robust Sparse Regression**

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zhang, Wangyang; Balakrishnan, Karthik; Li, Xin; Boning, Duane S.; Saxena, Sharad; Strojwas, Andrzej; Rutenbar, Rob A.

被引用次数: 18

2013

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**Large-area and high-resolution distortion measurement based on moire fringe method for hot embossing process**

OPTICS EXPRESS

Xu, Zhiguang; Taylor, Hayden K.; Boning, Duane S.; Yoon, Soon Fatt; Youcef-Toumi, Kamal

被引用次数: 9

2009

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**Three-dimensional profile stitching based on the fiducial markers for microfluidic devices**

OPTICS COMMUNICATIONS

Xu, Zhiguang; Li, Shiguang; Burn, Daniel J.; Shilpiekandula, Vijay; Taylor, Hayden K.; Yoon, Soon Fatt; Youcef-Toumi, Kamal; Reading, Ivan; Fang, Zhongping; Zhao, Jianhong; Boning, Duane S.

被引用次数： 11 2009

Study of CMOS Process Variation by Multiplexing Analog Characteristics

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Gettings, Karen M. G. V.; Boning, Duane S.

被引用次数： 4 2008

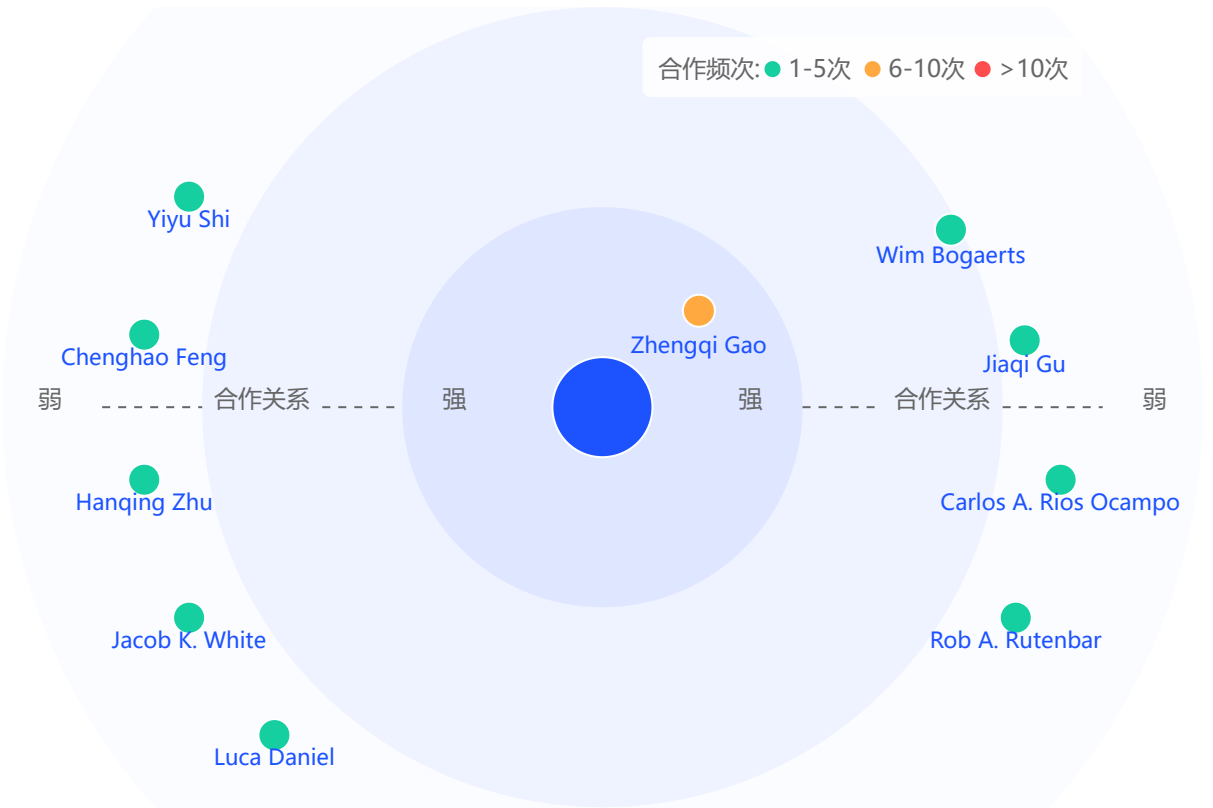
Variation

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Boning, Duane S.; Balakrishnan, Karthik; Cai, Hong; Drego, Nigel; Farahanchi, Ali; Gettings, Karen M.; Lim, Daihyun; Somani, Ajay; Taylor, Hayden; Truque, Daniel; Xie, Xiaolin

被引用次数： 29 2008

学术合作



Zhengqi Gao 合作次数： 10

麻省理工学院

Wim Bogaerts 合作次数： 3

根特大学

Jiaqi Gu 合作次数： 2

亚利桑那州立大学

Carlos A. Rios Ocampo

马里兰大学-大学城

合作次数: 1

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Rob A. Rutenbar

伊利诺伊大学厄巴纳-香槟

合作次数: 1

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还有 5 个合作者